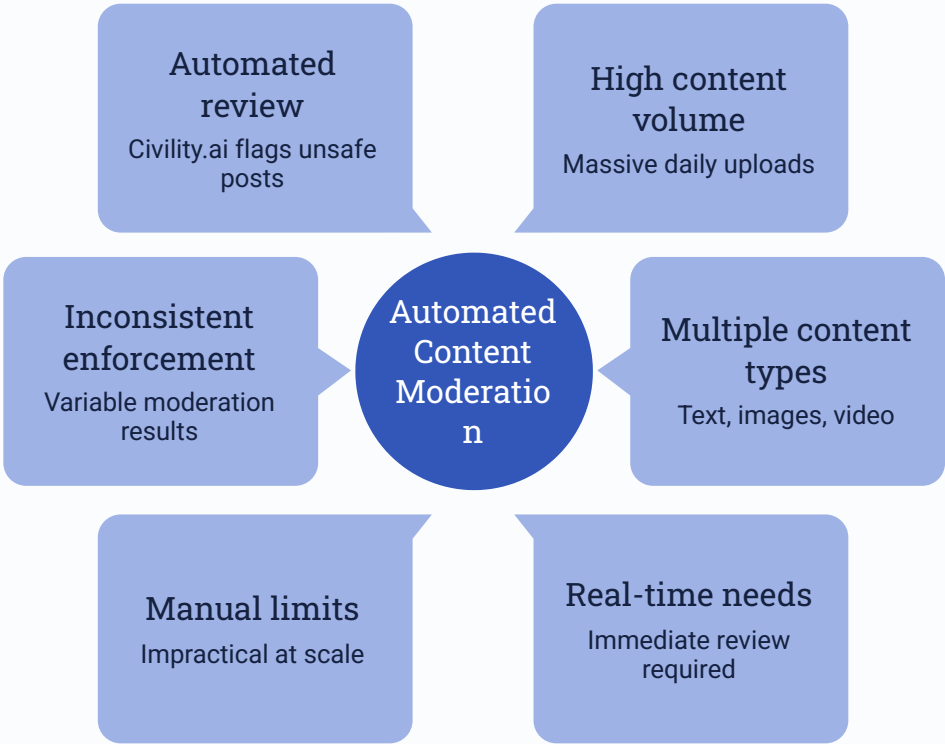


Problem Understanding

Social media platforms receive enormous volumes of user-generated content daily, including text posts, images, and videos. Without automated moderation systems, harmful, abusive, violent, or inappropriate content may be published and spread rapidly, negatively impacting users and platform safety.

Civility.ai addresses this by automatically reviewing uploaded content before publication, ensuring that only safe content reaches the platform whilst flagging potentially harmful material for appropriate action.



Solution Workflow

The system follows a real-time moderation pipeline that processes content through multiple stages to ensure comprehensive analysis and appropriate action.



Abuse Score Logic

The system generates an abuse score from 1 to 100 for each piece of content, determining the appropriate moderation action based on severity levels.

 0–25: Safe Content is clean and appropriate. Published immediately with no restrictions.	 26–50: Warning Moderate concerns detected. Content may be flagged for manual review or age-restricted.
 51–75: Moderate Abuse Significant violations identified. Content blocked and user receives warning. History updated.	 76–100: Severe Abuse Critical violations detected. Immediate blocking, user suspension initiated, admin alerted.

High-Level Architecture

Technology Stack

- **Frontend Framework:** Css, React
- **Backend Framework:** Django for robust server-side processing
- **Database:** MongoDB for flexible document storage
- **AI Moderation:** Gemini LLM for content analysis
- **Email System:** Gmail SMTP for admin alerts

Key Design Principles

- Real-time processing with low latency
- Scalable architecture for high traffic
- Secure data handling and privacy
- Extensible design for future features

Architecture Flow

User request reaches backend → moderation engine processes content → score generated → stored in database → action returned to platform

Key Features

Unified Moderation Pipeline Single AI moderation pipeline processes text, image, and video content types, ensuring consistent enforcement across all user-generated content	Repeat Offender Detection Tracks user history and cumulative abuse scores to identify patterns of harmful behaviour and escalate repeat violations appropriately	Admin Alert Escalation Automatically notifies administrators only for high-risk cases (flagged_count ≥ 3, abuse_score > 50, or severe abuse detected) with detailed reports
Abuse Score-Based Decisions Dynamic decision-making based on 1-100 abuse score range with configurable thresholds for different content types and platform policies	Multilingual Support Current support for English, Hindi, Telugu, Tamil, and mixed-language content improving accuracy for regional users	

Challenges & Innovation

Key Challenges Addressed

- **Multilingual Abuse Understanding:** Regional dialects and context-specific language patterns requiring sophisticated NLP models
- **Regional Slang Detection:** Platform-specific and cultural variations in harmful language that vary by geography
- **Video Frame Processing:** High computational cost of analysing multiple frames per second with acceptable latency